# SiteGuard

PPE compliance detection for construction sites: final project report

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## 1. What it does

SiteGuard looks at a photo, a video or a live camera, finds every worker, decides for **each worker separately** whether they wear the required equipment, and says exactly what is missing. A helmet anywhere in the picture never makes everyone compliant. When it is not sure, it says "needs review" instead of accusing anyone. Anyone who broke a rule is sent to a supervisor with the picture, the place, the time and an auto-written remark; the employee sees it in their own feed; and a person decides before anything counts as a violation or a penalty.

### Problem statement requirements

| Requirement | How it is met | Status |
|---|---|---|
| Detect each worker | YOLO11 person detector, plus zoomed tiles for small or distant workers. | Met |
| Decide per worker whether the required PPE is worn | Each helmet and vest is assigned to exactly one worker by position (head zone, torso zone), so one helmet in the picture cannot cover everyone. | Met |
| Flag violations and the missing item | Verdicts COMPLIANT, VIOLATION (with the missing item) or NEEDS REVIEW. | Met |
| Annotated images with a verdict per worker | Boxes and labels on every picture, in the demo desk and in each case. | Met |
| Violation summary | Supervisor overview, cases table, employee pages, PDF reports. | Met |
| Multiple workers, occlusion, distance, poor lighting | Zoomed tiles, dark-photo enhancement, weak-evidence rule, edge-cut checks. Hard cases still fail sometimes (section 5). | Met, with limits |

## 2. Architecture

**Input**
Photo, video or live camera on the demo desk (port 3003).

→

**Detector service (Python, port 5000)**
People, helmets, vests, heads. Object check. Per-worker rules. Second-model cross-check. Naming.

→

**API (Node, port 4000)**
Cases, one violation per person per day, remarks, penalties, salary, alerts.

→

**Two portals**
Supervisor panel (3002). Employee portal (3001).

| Part | Technology | Job |
|---|---|---|
| Detector | Python 3.12, Ultralytics YOLO11, PyTorch, OpenCV | Finds people and gear, applies the per-worker rules, tracks people in video. |
| Object check | YOLO11 pose model and the COCO person detector | A box that holds no body (for example a concrete bucket on a crane) is marked "OBJECT: not a worker" and never flagged. |
| Cross-check | Second YOLO model (extras.pt) | If it sees a helmet on someone the main model says has none, the call is held back for review. |
| Naming people | OpenCV YuNet (faces) and SFace (recognition) | Face first; then a strict dress-colour match; otherwise a made-up name tied to that footage. |
| API and database | Node.js, Express, NeDB (MongoDB-style, embedded) | System of record; swappable for MongoDB by changing one file. |
| Websites | React (Vite), three separate sites | Supervisor panel, employee portal, demo desk with spoken alerts (browser speech). |

### How one catch flows

- The demo desk sends the picture or video to the detector. Everyone is checked; "no helmet" calls are cross-checked.
- Each person who broke a rule gets a name: an enrolled employee if their face is known, otherwise a made-up name that is the same every time for the same footage. Only those people are spoken to, at most once every 10 minutes.
- The case goes to the supervisor automatically with its picture and a remark written from the facts (who, what, where, when, how many times, earlier violations). The supervisor can edit the remark.
- The employee sees it at once in their personal feed (date, time, place, how it was caught). It becomes a violation only when the supervisor confirms it.
- **One violation per person per day.** Caught again the same day: only the catch count goes up. The next day: a new violation. Three confirmed violations in 30 days propose a penalty, which a supervisor must approve and the employee can dispute.

## 3. Data, models and licences

| Item | Source | Licence |
|---|---|---|
| Training photos | Construction-PPE (Ultralytics), 1,416 photos; CHVG (Ferdous and Ahsan 2022), 1,699 photos | AGPL-3.0; CC BY 4.0 |
| Detection model | YOLO11 (Ultralytics), fine-tuned by us to person, helmet, vest and head | AGPL-3.0 |
| Face models | OpenCV YuNet and SFace (OpenCV Zoo) | MIT; Apache-2.0 |
| Fall classifier (experimental) | GMDCSA24 dataset, 160 clips | MIT |

Credits and licence notices are kept in the README and docs/LICENSES_AND_SOURCES.md and are not removed. Nothing was copied from other projects; other repositories were read for ideas only.

## 4. Measured accuracy

> **We do not claim 99.99%.** No detection system can honestly promise that. These are the numbers measured on photos the model never saw, with scripts that can be re-run ( `scripts/compare_models.py` , `scripts/eval_workers.py` , `scripts/eval_crosscheck.py` ). Full detail: `docs/ACCURACY_MEASURED.md` .

| What | Result |
|---|---|
| Detection score on 314 held-out photos (AP at IoU 0.5) | person 0.925, helmet 0.941, vest 0.901, head 0.736; **mean 0.876** |
| "No helmet" verdicts on 661 labelled workers (60 px or taller) | caught 78, false alarms 7, missed 4 (14 more sent to review): **precision 0.918**, **recall 0.812** (0.958 counting "send to review") |
| Dark photos (darkened copies of the test photos) | night model: "no helmet" precision **0.890** (original model alone 0.647); 18.9% of workers not found |
| Missing vest (101 photos, 78 workers without one) | 54 caught automatically (69%), 17 more sent to review; 6 of 89 vest-wearers wrongly flagged |
| Workers the detector never found | 85 of 661 (12.9%) |
| Vest false alarms on people who do wear one | 14 of 312 (4.5%) |
| Second-model cross-check on helmet calls | held back 0 of 88 real violations, caught 2 of 6 false alarms (a small sample) |
| Same person by face (160 clips, 4 people) | no two people merged; 64 of 64 named observations correct; each person ended up with 1 or 2 names; about 20% of faces left unnamed rather than guessed |
| Same person by clothes | weak: at a setting that wrongly links people 0.14% of the time, only 4% of same-person pairs are recognised; used only for the same footage seen again |
| Tried and rejected | a larger model (0.819, lower), threshold tuning, test-time augmentation (lower), an appearance-based tracker (no gain) |
| Fall alerts (experimental, off by default) | per clip: 53% of falls caught (24% in low light), false alarm on 7% of normal clips; weak |

## 5. Limits, privacy and next steps

### Where it still fails

- About 13% of labelled workers are never found (small, hidden or in a crowd).
- Hi-vis shirts and life jackets can pass as vests (no training photos of them yet).
- Backlit or silhouetted workers can be wrongly flagged; they should be "needs review".
- Faces narrower than about 48 pixels (typical in wide crowd shots) cannot be recognised, so those workers get a made-up name rather than a real one.
- Not yet tested on real footage: the live camera and the spoken alerts. Goggles, gloves and boots use a second model whose accuracy has not been measured.

### Privacy and fairness

- A catch never becomes a violation or a penalty without a supervisor's decision. The system only proposes penalties.
- Face recognition needs the employee's agreement; only a numeric template is stored, never the photo. Made-up names belong to nobody and expire after 7 days. For real use, switch them off ( `DUMMY_NAMES=0` ). See docs/PRIVACY_NOTES.md (India's DPDP Act 2023).
- Salaries, names and penalty amounts in the demo are made up. Whether wages can lawfully be reduced for a penalty is for HR and the company to confirm.

### Next steps

- Hard-negative photos (hi-vis shirts, life jackets) and more small, distant workers; then retrain and re-measure.
- Test the live camera and spoken alerts on real site footage.
- Real accounts and passwords, consent and data retention, then host it (the three sites on a static host such as Vercel; the API and detector on a server).

## 6. How to run it

| Step | Command or address |
|---|---|
| Start everything | `npm run start:all` in `siteguard-web` (API, three sites and the detector service) |
| Demo history (optional) | `node api/src/seed.js --demo` |
| Check it all works | `npm run smoke` and `npm test` ; in `ppe-detection` : `python -m pytest` |
| Supervisor panel | http://localhost:3002 sign in `S001` / `S001@123` |
| Employee portal | http://localhost:3001 sign in `E001` / `E001@123` (the password is the code followed by @123; E001 to E120) |
| Demo desk | http://localhost:3003 (no sign-in; click "Turn on voice alerts" first) |

## 7. Questions judges may ask

### How accurate is it?

On 314 photos it never saw, the mean detection score is 0.876. For "no helmet" on 661 labelled workers, 91.8% of the people we flag really have none, and we flag or send to review 95.8% of those who do. We do not claim 99.99%; about 13% of workers are not found at all, and we say so.

### What if a helmet is in the picture but not on that person?

Each helmet is assigned to exactly one worker by where it sits on the head. A helmet on one person, in a hand, or on a table does not cover anyone else. The zone rules and the "needs review" outcome are built for this.

### Why not just flag everyone it is unsure about?

Accusing a person on weak evidence is unfair. Weak evidence goes to "needs review", and a second model double-checks every helmet call before it is sent on.

### Can it punish someone wrongly?

It cannot punish anyone by itself. The system only proposes a penalty after three confirmed violations in 30 days; a supervisor must approve it, the employee can dispute it, and the amounts are examples that management sets.

### How do you know who the person is if their face is hidden?

When the face is clear (48 px or wider) we match it to an enrolled employee, with consent. Clothes are a weak signal and used only under a very strict rule. Otherwise the person gets a made-up name for that footage, and the screens say the system does not know who they really are.

### What stops the same person being punished many times in a day?

One violation per person per day. A repeat catch the same day only raises the catch count; the next day a new violation can be created.

### What about privacy?

Only a numeric face template is stored, never the photo, and only after consent. Made-up identities expire after 7 days and can be cleared. A privacy note maps this to India's DPDP Act; real use needs a legal review.

### What did you build yourselves, and what did you reuse?

The per-worker matching rules, the verdict logic, the object check, the cross-check, the penalty policy, the three websites and the API are ours. The detection model is a fine-tuned open-source YOLO11 and the face models are OpenCV's; all credited and licensed.

### Does it work on live CCTV?

There is a live camera mode and a CCTV script that tracks people and sends violations to the supervisor. It is tested on video files and simulated sequences; we have not yet tested it on real site footage, and we say that openly.

### What would you do with more time?

Collect hard-negative photos (hi-vis shirts, life jackets) and more small, distant workers, retrain and re-measure; test on real footage; add real accounts, consent and data-retention rules; and host it.